

QAARSUT UUMMANNAQ, Greenland

WMO: 042130

Lat: 70.733N

Lon: 52.700W

Elev: 89

StdP: 100.26

Time Zone: -3.00 (GRL)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-26.7	-24.4	-33.7	0.2	-25.2	-31.1	0.2	-22.8	15.6	-8.6	14.0	-8.6	3.0	140	0.958

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	16.1	7.9	14.6	7.4	13.3	7.0	9.6	11.7	8.8	11.3	8.3	11.1	8.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.1	9.5	8.0	6.7	8.8	7.2	6.4	8.3	28.4	11.6	26.9	11.3	25.6	11.2	12.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.8	11.4	DB	-27.0	20.0	3.3	5.2	-29.4	23.7	-31.4	26.7	-33.2	29.7	-35.6	33.4	(4)
(5)			WB	-27.0	10.7	3.4	0.7	-29.4	11.2	-31.4	11.6	-33.3	12.1	-35.7	12.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-11.8	-13.5	-13.6	-8.0	-0.4	6.0	9.3	7.9	2.5	-3.0	-6.9	-9.9	(6)	
	DBStd	9.37	5.89	7.06	6.71	6.26	4.12	3.49	3.13	2.47	2.86	3.92	4.62	4.94	(7)	
	HDD10.0	4927	675	659	731	540	323	126	50	74	224	403	507	616	(8)	
	HDD18.3	7923	933	892	989	789	581	369	279	324	474	662	757	874	(9)	
	CDD10.0	47	0	0	0	1	1	7	30	8	0	0	0	0	(10)	
	CDD18.3	1	0	0	0	1	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	13	0	0	0	13	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	6	0	0	0	6	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.6	5.1	4.6	3.7	3.8	4.1	4.3	4.4	4.3	4.4	5.0	5.8	5.6	(14)	
(15) Precipitation	PrecAvg	432	39	25	22	24	23	24	29	46	44	48	54	54	(15)	
	PrecMax	641	141	55	76	68	69	68	70	96	99	113	137	110	(16)	
	PrecMin	222	2	2	3	1	3	2	3	14	9	12	5	8	(17)	
	PrecStd	107	28	17	18	15	15	16	16	23	20	25	28	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	5.4	5.3	23.3	13.6	16.3	18.2	16.5	10.6	8.7	7.1	4.5	(19)	
		MCWB	0.8	0.4	1.8	8.4	6.4	7.5	8.9	8.2	4.5	2.9	2.9	0.6	(20)	
	2%	DB	1.1	1.0	1.3	7.5	9.8	14.5	16.7	14.3	9.1	6.3	4.4	1.3	(21)	
		MCWB	-1.9	-2.8	-1.3	3.0	4.8	7.2	8.3	6.9	4.1	1.5	0.6	-1.2	(22)	
	5%	DB	-0.9	-0.8	-1.6	3.8	7.5	13.2	15.3	13.1	7.7	4.0	2.6	-0.9	(23)	
		MCWB	-3.5	-3.7	-4.0	0.1	3.4	6.8	7.7	6.6	3.3	0.6	-0.7	-3.2	(24)	
	10%	DB	-3.1	-2.8	-4.0	0.3	5.2	11.7	14.0	11.9	6.3	2.0	0.2	-3.3	(25)	
		MCWB	-5.3	-5.3	-5.8	-2.1	1.9	6.3	7.4	6.4	2.8	-0.4	-2.0	-5.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	0.9	1.9	4.7	7.1	9.2	10.9	10.1	6.7	4.1	3.5	1.1	(27)	
		MCDB	3.3	5.1	4.9	9.5	11.2	12.2	13.3	11.6	8.6	6.7	6.1	3.6	(28)	
	2%	WB	-1.3	-2.3	-1.2	2.0	5.1	8.4	9.9	9.1	5.1	2.1	1.2	-0.8	(29)	
		MCDB	0.6	0.3	1.0	5.7	8.6	12.0	12.1	10.7	6.7	4.9	3.9	1.2	(30)	
	5%	WB	-3.4	-3.6	-3.8	-0.4	3.6	7.6	9.2	8.4	4.4	1.0	-0.5	-3.1	(31)	
		MCDB	-1.0	-0.8	-1.9	2.6	6.7	11.4	11.7	10.3	6.2	3.5	2.0	-1.1	(32)	
	10%	WB	-5.2	-5.4	-5.8	-2.2	2.5	6.8	8.7	7.8	3.7	-0.1	-2.0	-5.0	(33)	
		MCDB	-3.1	-2.7	-4.1	-0.2	4.8	10.6	11.5	10.0	5.7	1.8	0.4	-3.4	(34)	
(35) Mean Daily Temperature Range	MDBR		4.6	5.2	5.7	7.2	5.2	5.3	5.3	4.9	3.9	3.6	3.8	3.8	(35)	
	5% DB	MCDBR	7.0	7.3	7.9	14.9	8.5	7.8	7.7	7.2	5.3	5.7	6.4	5.9	(36)	
		MCWBR	5.3	5.2	5.9	5.9	4.5	3.0	2.9	3.2	3.3	3.7	4.7	4.8	(37)	
	5% WB	MCDBR	7.1	7.4	7.2	8.5	7.5	6.5	5.7	5.3	4.3	4.9	6.5	5.7	(38)	
		MCWBR	5.5	5.7	5.7	6.0	4.6	3.2	3.2	3.4	3.5	3.7	5.1	5.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.212	0.257	0.326	0.341	0.355	0.351	0.334	0.290	0.221	N/A	N/A	(40)	
	taud		N/A	1.964	2.142	1.987	2.110	2.266	2.346	2.387	2.340	2.009	N/A	N/A	(41)	
	Ebn at Noon		0	471	741	777	824	828	819	788	709	433	0	0	(42)	
	Edh at Noon		0	39	73	121	123	109	98	83	62	34	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.00	0.30	1.75	4.05	5.66	6.11	5.59	3.70	1.89	0.54	0.00	0.00	(44)	
	RadStd		0.00	0.04	0.13	0.32	0.37	0.41	0.51	0.32	0.15	0.04	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.87	+3.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page